

CLOUD COMPUTING

Project: Development of a Cloud-Based Video Fingerprint Collection System for an Insurance Company

Context

An insurance company aims to implement a **cloud-based data collection system** for its subscribed drivers. Each driver is equipped with a surveillance camera installed on the vehicle's windshield, functioning as a *black box*. These cameras allow drivers to submit video evidence in the event of an accident.

To reduce the storage burden and enable scalable analysis, the company does **not** require raw video uploads. Instead, only **video frame fingerprints**—unique cryptographic hashes—must be generated and sent to the cloud in real time. These fingerprints will serve as a reference to **authenticate** any future video submission after an incident.

The system may rely on a **smartphone mounted to the windshield**, which captures the road ahead and generates hashes of each video frame. These hashes must be **continuously transmitted to a cloud backend**, emphasizing the use of **cloud computing platforms** such as **Supabase**, **Firebase**, or any modern serverless backend.

This topic places strong emphasis on **principles and benefits of cloud computing** such as real-time data handling, data availability and secure data storage.

☛ Cloud Computing Requirements

Your proof-of-concept application must simulate the behavior of the final system while making real use of open-source **cloud-based infrastructure**. You are free to choose your technology stack (web, mobile, or desktop) and programming language, but your solution must demonstrate real-time or near-real-time transmission of data to a selected cloud storage platform (e.g., Supabase, Firebase Firestore) or a close simulation.

✂ Technical Requirements

1. Input Handling:

The application must read a .txt file containing a list of generated fingerprints (hashes of video frames or texts), one per line by following secure standards (e.g., **SHA-256** or **SHA-512**).

2. **Timestamping:**
Each fingerprint should be associated with a **timestamp**.
 3. **Real-Time Transmission to the Cloud:**
Simulate real-time transmission of each fingerprint to your chosen open source cloud platform or database.
 4. **Cloud Storage Integration:**
The data must be persistently and securely stored in a kind of **cloud-hosted database** demonstrating cloud computing usage in practice.
 5. **Code Reuse & Attribution:**
You may use external libraries or open-source code, provided that **you clearly cite all sources** in your documentation and comments. Plagiarism is prohibited.
 6. **Validation of data integrity** from the insurance side.
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Evaluation Criteria (Total: 20 points)

Criteria	Details	Points
1. Application Architecture	Modular, well-organized structure adapted to the chosen platform (web, desktop, or mobile).	1
2. Input File Handling	Correct processing of .txt file with fingerprints.	2
3. Real-Time Streaming Simulation and validation of data integrity	Fingerprints transmitted in a simulated real-time flow.	4
4. Data Format and Security	Use of e.g. SHA-256/512, timestamping and secure data transmission.	1
5. Code Quality and Reuse Transparency	Clean, documented code. Reused code or libraries are clearly cited and integrated properly.	1
6. Documentation (README)	Clear setup guide, architecture overview, technology choices explained.	1
7. Final Submission (Demo video and codes)	A video demonstration of the application (e.g. from a windows screen recorder tool), along with full source code, must be submitted via email to the address provided above.	10